

Greta 2.0: Social Interactive Agent system, optimized for neural network integration

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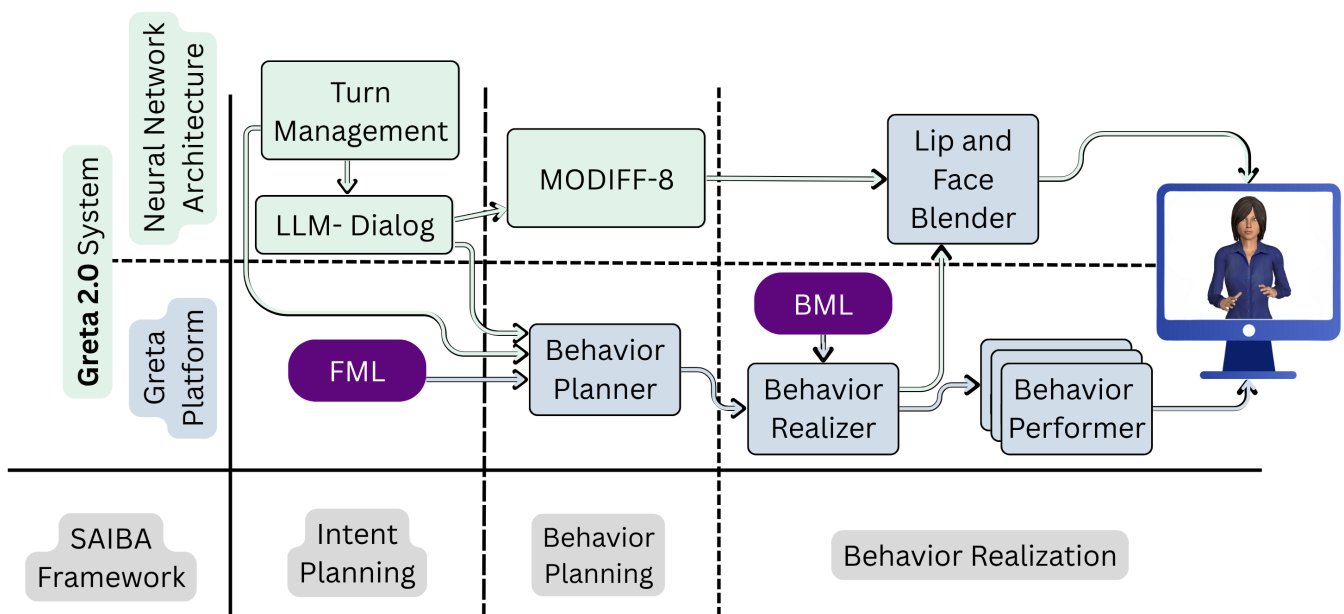


Figure 1: Greta 2.0 system architecture: Green color indicates added modules to the Greta platform, blue indicates existing modules of the Greta platform following the SAIBA framework, gray indicates abstract concepts, and purple indicates types of XML files that exchange information between modules. Each new module in green is explained and described in the paper.

Abstract

Social interaction is a multimodal phenomenon, including various verbal and nonverbal behaviors that are time-synchronized across different interaction partners. Socially Interactive Agents (SIAs) have been developed as interaction partners for humans. With the emergence of large language models, human-SIA interactions

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have become increasingly similar to human-human interactions. This paper proposes an upgraded SIA system, Greta 2.0, which can work with conventional cascaded-module systems and frame-by-frame behavior generation systems based on neural networks. To demonstrate Greta 2.0’s capabilities, we integrated ASR, LLM, a facial expression generation model, and a turn-taking prediction model to achieve lifelike SIA. The source code, demo videos, and documentation are publicly available at our project page <https://github.com/isir/greta>.

CCS Concepts

• **Human-centered computing** → *Interaction design; Collaborative interaction; HCI design and evaluation methods; HCI theory, concepts and models*; • **Computing methodologies** → *Artificial intelligence*.

Keywords

Socially Interactive Agent, Virtual Agent, Social Robot, LLM, ASR, Turn Taking, Facial Expression Generation, Real-time Diffusion

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1 Introduction

A Socially Interactive Agent (SIA) is a computer-generated virtual character capable of interacting with humans [25, 34]. Unlike voice assistants like Siri from Apple or Alexa from Amazon, SIA can communicate through facial and body gestures, making interactions more expressive and closer to those between humans.

The recent technical evolution of large language models based on neural networks (e.g., ChatGPT, Mistral) has drastically changed the development of human-SIA interactions. These large language models started from textual interaction and expanded to multimodal interaction, considering other information modalities simultaneously. Although several dialog and behavior generation models [8, 22] have been proposed, only a few have been implemented into SIA platforms. One of the implementation bottlenecks is on the platform's core architecture. Some early SIA platforms were designed based on the cascaded behavior realization architecture, known as SAIBA [20]. However, since it processes data in a one-way structure, generating adaptive behaviors in real-time is challenging.

This paper introduces an integrated SIA system, Greta 2.0, on the Greta platform that works with neural network models. The Greta 2.0 system integrates models for generating adaptive behavior, dialog, and turn-taking mechanisms. To overcome the difficulty of generating adaptive behavior, the system uses two independent pathways: one from the SAIBA process and another from a frame-by-frame generative neural network, as depicted in Figure 1. The latter pathway enables the computation of adaptive behaviors frame by frame in real-time. It takes the extracted user behaviors and the agent behaviors displayed so far as input to compute the agent behavior in the next frame. The behaviors generated by these processes, SAIBA and neural networks, are synchronized and blended before rendering the animations. These parallel executions minimize response generation delay, ensuring a real-time interaction with human participants. Furthermore, since we use a modular architecture, each functionality can be easily removed or added.

While commercial companies have developed products that allow for realistic virtual character interactions, researching and developing on top of them has been challenging because not all of their internal mechanisms are open-source. Therefore, we provide this system as open-source software, except for some external API calls for heavy computations. We have prepared the complete source code for the project on the project page <https://github.com/isir/greta>. The documentation and videos demonstrating the capabilities of the various modules can be found on the project's wiki page.

In the remainder of the paper, we present existing SIA platforms and integrated systems in Section 2. Then, we explain the core of our Greta 2.0 system in Section 3 and discuss newly developed

functionalities, including neural network integrations for dialog in Section 4, turn management in Section 5 and facial expression generation in Section 6. Finally, we conclude and discuss future possibilities in Section 7.

2 Related work

SAIBA (Situation, Agent, Intention, Behavior, Animation) architecture is the first unified architecture to control the synchronized verbal and nonverbal behavior of SIA characters [20]. It comprises three different behavior generation processes: planning a communicative intent, planning a multimodal realization of this intent, and realizing the planned behaviors as depicted at the bottom of Figure 1. Two XML-based behavior representation formats, called Functional Markup Language (FML) and Behavior Markup Language (BML), exchange messages between the three modules [17, 20]. Former SIA platforms, including Greta, followed this architecture.

One of the first-generation SIA platforms of SAIBA architecture is SmartBody [43]. This platform takes BML and audio signals as input and calculates the nonverbal behaviors of the SIA. It also developed a nonverbal behavior generator (NVBG) module that generates corresponding communicative gestures based on input text. The SmartBody synchronizes every modality based on audio and timing information from the speech synthesizer. Then, it passes those scheduled behaviors to the motion control engine, followed by animation rendering engines. Similarly, the Greta platform was proposed as a module-based, multimodal behavior generation system compliant with the SAIBA architecture, working in real-time [28]. It provides a GUI application called *modular* to control connections between modules, making it easier to customize for each project's specific requirements. The Greta platform has been extended to control a virtual agent with reciprocal facial expression generation for cognitive behavioral therapy and social skills training [39, 46]. AsapRealizer 2.0 is an extension of AsapRealizer and Elckerlyc [21, 44, 45]. While the last two platforms follow SAIBA's modular and uni-directional approach, AsapRealizer 2.0 employs an incremental realization approach for multimodal behaviors, based on an incremental text-to-speech (TTS) system, ensuring their coordination and synchronization. AsapRealizer 2.0 synthesizes audio incrementally and manages synchronized behaviors with bidirectional information flow, enabling reactive and adaptive behaviors. Based on this platform, the SIA Billie has been designed to accompany and guide elderly users throughout the day [19]. The SERA ecosystem has been developed by extending the SAIBA framework to incorporate users' behavioral perceptions [35]. It comprises reusable tools, namely: Thalamus, corresponding to the intent planner; Skene, for the behavior planner; and Nutty Tracks, for behavior realization following the SAIBA framework. By adding plugins for each agent or robot in the Nutty Tracks, it is possible to generate synchronized, frame-by-frame animations for virtual and robotic characters using the unified mechanism. These SIA platforms have been integrated into interaction systems, which are larger architectures that model various applications. For example, VHToolkit integrates SmartBody and various extension modules (e.g., dialog management, emotion model, multimodal perception, speech synthesizer) as a single package, which has been used in multiple projects such as MRE and SASO [15, 36, 42, 43]. The TARDIS is a scenario-based serious game

simulation system that supports social training and coaching with virtual agents in the context of job interviews, developed on top of the Greta agent platform [2]. It comprises four modules: a scenario module, a social signal module, an affective module, and an animation module. The TARDIS controls the SIA's actions based on context from the scenario module and affective states calculated from the social signal and the affective modules. These two interaction platforms are compatible with the ALMA layered affective model [13].

Although these academic SIA platforms have served flexible integrability, rapidly integrating up-to-date components is challenging due to workload limitations. Recently, to overcome this limitation, VHToolkit 2.0 was launched to integrate legacy standalone software and recent API-based web applications, providing easily affordable yet flexible SIA systems [14]. By implementing interfaces to commercial products, a combined system can be developed that remains open and flexible while offering the superior reliability and robustness of commercial software. Members of the spoken-dialog community have employed this approach and developed MMDAgent-EX and Remdis [6, 24]. There are also commercial SIA platforms and interaction systems. For example, the Nvidia Digital Humans platform integrates Riva natural language understanding modules, Nemotron language models, and audio-to-face conversion models, among others [29]. Since our focus is on interactions, whereas their focus is on speech and dialog, the interested readers can refer to the cited papers for more details.

Most existing platforms use either SAIBA-based cascaded behavior generation or frame-by-frame behavior generation based on a diffusion neural network, but cannot do both. The system presented in this paper can combine the two approaches; an example of this combination is our facial expression generation (MODIFF-8) model (see Section 6). Furthermore, our system leverages pretrained neural networks to realize smooth spoken dialog interactions, specifically for ASR, LLM, and turn-taking management based on voice activity projection (VAP) models.

3 Architecture: two pathways synchronization

In this section, we present the core of the Greta 2.0 system that supports two pathways: the classical SAIBA framework and a frame-by-frame neural network-based generation. Figure 1 shows the primary process of the Greta 2.0 system. In the Greta 2.0 system, one possible pathway follows the SAIBA framework. A sequence of communicative intentions in the APML-FML format [26], an extension of Functional Markup Language [17] with Affective Presentation Markup Language [5], is sent to the Greta 2.0 system as input to the Intent Planner module. This sequence is then sent to the Behavior Planner module, which outputs sequences of multimodal behaviors encoded using the Behavior Markup Language (BML) [20]. Finally, the Behavior Realizer module computes the animations of the agent in terms of Facial Animation Parameters (FAPs) and Body Animation Parameters (BAPs) [16, 32]. The alignment of the text to be said and the display of communicative intentions is done by positioning timestamps.

With the emergence of neural networks, frame-by-frame generative models have significantly improved the quality of multimodal behavior in terms of rhythm and expressiveness. Although SAIBA

architecture is well-designed to generate multimodal behaviors from text input, it is not adapted for these latest computational approaches, nor for modeling fast adaptive mechanisms or incremental models. To integrate both computational approaches, the Greta 2.0 system features a bypass pathway that processes the generated frame-by-frame face/body behaviors via FAP frames and BAP frames, which are frame-level representations of the agent animation, as depicted in the second line of Figure 1. To blend the two FAP signals, which originate from different pathways (one from the SAIBA process and the other from the frame-by-frame neural network), we implemented two additional modules: a face blender and a lip blender. Since each FAP/BAP frame contains timestamp information, we can identify frames with the same timestamp coming from different process paths and blend them accordingly.

The core part of the Greta 2.0 system is implemented in Java. However, most recent state-of-the-art neural networks, including our new modules, are implemented in Python. Therefore, the system synchronizes frame-level timestamps between the Java and Python systems. The timestamp in the Java system is defined as the elapsed time from the system's launch. Similarly, the Python system also adds the elapsed time from the launch of the Python system for each behavior frame. Once the frames are sent from the Python system to the Java system, the Java system adds the current timestamp to the timestamps of the frames generated by Python, synchronizing the frames between the two systems. Figure 2 shows the detailed architecture of the Greta 2.0 system with the data flow. In the latter part of this paper, we will detail the three main modules, shown in green in the Figure, in the subsequent sections.

4 LLM/ASR Integration

To enhance Greta 2.0's conversational capabilities, we integrated two new modules: ASR and LLM-Dialog. These modules enable fluid interaction with the SIA by leveraging the generative power of large language models to create a dialog agent from a simple prompt. For more fine-grained control over the conversation flow, it is also possible to use a dialog manager and a natural language understanding module. The integration provides a simple interface, as illustrated in Figure 3. The user can enter a prompt in the System prompt text box and interact with the LLM through the Greta agent without any additional coding. Alternatively, the prompt can be selected automatically by the dialog manager.

4.1 ASR module

The ASR module interfaces with the Greta 2.0 system and external APIs that convert microphone input into text transcripts. Currently, the module supports only the Deepgram API, which utilizes large, deep learning models for speech-to-text tasks. However, the implementation, being modular, is designed to be easily adapted to other ASR providers. Currently, the module supports both French and English, offering flexibility for multilingual integration. When not used in conjunction with the Turn Management module (see Section 5), the ASR module can operate in three distinct modes:

- **Continuous Mode:** The audio input module, through a microphone, continuously listens to the user, capturing their speech. This mode should only be used with headphones. Otherwise, the system might also pick up the agent's speech.

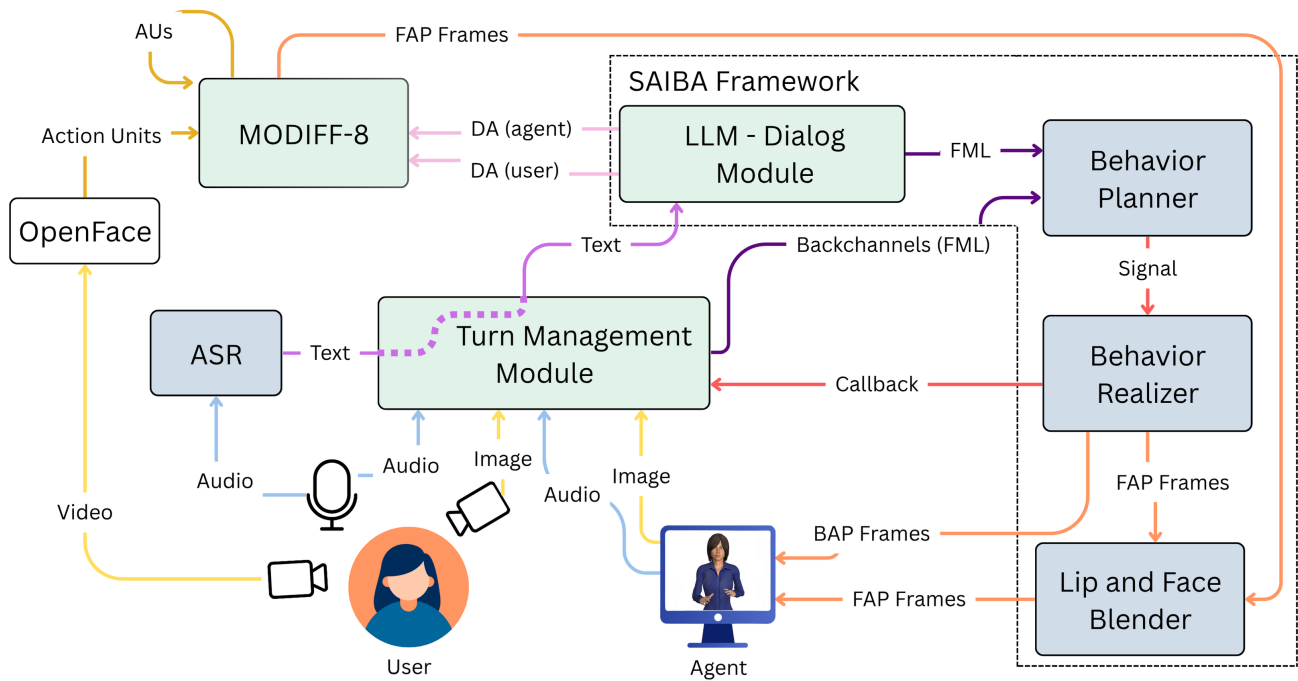


Figure 2: Greta 2.0 detailed system architecture: The colors of the modules have the same meaning as in Figure 1. In the paper, we focus on green-colored modules. While the Facial Expression Generator and Automatic Speech Recognition (ASR) work in parallel asynchronously, the Turn Management module controls the timing of backchannel generation and the response sentence generation via the LLM-Dialog Module. The LLM-Dialog Module also generates the Dialog Acts (DAs) sent to the Facial Expression Generator for the adaptive facial expression generation. The lines with different colors correspond to different data flows. The yellow and the blue lines correspond to user visual and audio data; the orange lines are animation data frames for the agent animation; the light purple line corresponds to transcribed text from the user’s audio; the dark purple lines are for the agent’s intents (both as speaker and as listener); the red lines are for the multimodal signals and callback signals.

This is a limitation of this mode; the other modes can be used without headphones.

- **Automatic Listening Mode:** This mode integrates simple turn-taking by connecting the Greta 2.0 Callback module to the ASR module. The user's microphone audio input is activated only when the agent is not speaking, thereby avoiding the capture of the agent's speech. The Callback module signals when the agent starts and stops talking, ensuring that the system listens when the agent is not speaking. In this mode, one limitation is that user interruptions are not considered. One of the other modes should be used for an application where interruptions are necessary.
- **Press-to-Talk Mode:** In this mode, the user (or experimenter when the system is used in experimental studies) must manually press the "Listen" button to activate the microphone (see Figure 3). This allows for complete control over the timing of turn-taking in the conversation.

The ASR module sends the transcribed text to the LLM-dialog module (presented in Section 4.2) once the end of the user’s speech is detected, i.e., after a silence of a particular duration in the user’s speech. If not used with the Turn management module (described in Section 5), the end of the user’s turn is determined by the Deepgram API, which detects silence for a predefined duration. This threshold

duration is set to 300 milliseconds by default, but this value is modifiable. This threshold value was chosen based on a test with individuals speaking French. However, appropriate values can vary depending on the language and situation, as the distribution of between-turn pauses can vary across languages [41].

4.2 LLM-Dialog Module

The LLM-Dialog module enables Greta 2.0 to generate responses with an LLM. It is connected to the ASR module to receive transcribed user input and respond accordingly. This module is composed of four sub-components responsible for managing different stages of the dialog process: **Natural Language Understanding (NLU)** module, **Dialog Manager**, **Language Generation** module, and **FML Converter**.

A detailed view of the data flow within this module is presented in Figure 4.

The NLU module analyzes the user’s utterance to identify the underlying intent (e.g., "Share personal information"). Based on this analysis, the Dialog Manager selects the appropriate system response as a dialog act. In the example in Figure 4, the dialog manager selects the dialog act "Ask about current emotion". The Language Generation module produces a textual response corresponding to the selected dialog act. Finally, the generated utterance

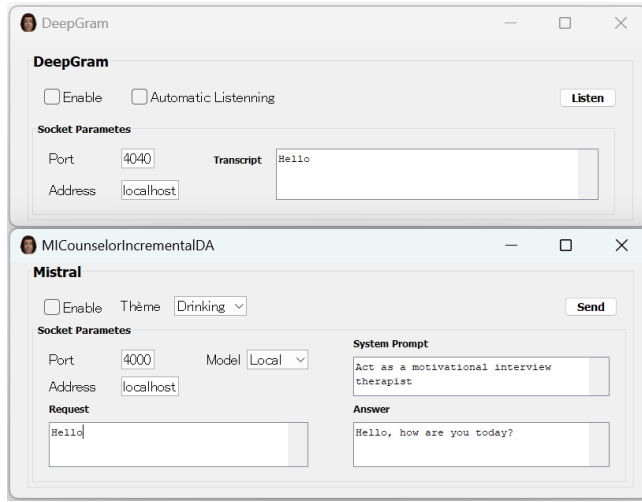


Figure 3: Interface of the LLM and ASR modules: "Listen" button controls activation/deactivation of the DeepGram ASR module, while the transcript is shown in the text box. LLM models can be selected from the selection box of the Mistral LLM module. The LLM response is generated and displayed in the text box in the "Answer" section based on the input text in the "Request" and "System prompt" sections.

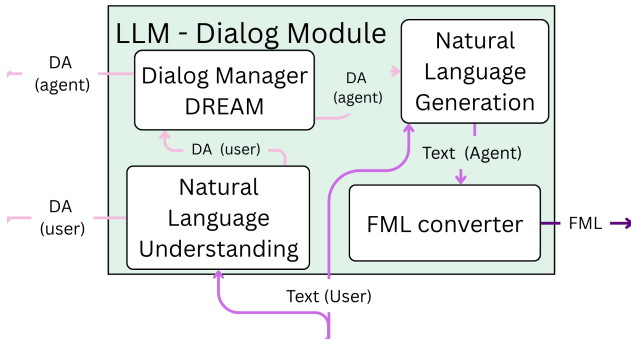


Figure 4: LLM-dialog module dataflow: Dialog Acts (DAs) are computed from the user's transcript and served as input to generate the agent text. The agent text is converted to an XML file in Functional Markup Language format. The DAs are also sent to the Facial Expression Generator to generate adaptive facial expressions for the agent. The pink lines represent the agent's and user's dialog acts; the purple lines represent the user's and the agent's text.

is transformed into FML by the FML Converter and sent to the Behavior Planner for execution. Although the NLU and Dialog Manager components are specifically designed for the Motivational Interviewing (MI) use case [10], these modules can be adapted for other applications. Meanwhile, the Language Generation module can also be used independently for simpler dialog scenarios.

In the following sections, we describe the three submodules: Language Generation, Natural Language Understanding (NLU), and the Dialog Manager.

4.2.1 Language Generation: LLM Module. The Language Generation module leverages a Large Language Model (LLM) to generate contextually appropriate sentences based on the user's utterances and a given prompt [10]. The system prompt can be customized through the Modular interface (see Figure 3) to define a specific role for the agent. For instance, to simulate a Motivational Interviewing (MI) therapist, the prompt could be: "Act as an MI therapist. Here are some instructions: ...". The module supports French and English interactions and is consistently guided to use conversational language, with the instruction: "You are a virtual assistant. Answer with short, oral-style responses." To ensure continuity and coherence across turns, the module includes the dialog history in each generation prompt. For faster response speed, the LLM's response is generated incrementally. The first sentence of the LLM's response is generated and sent immediately to the Behavior Planner as FML. Meanwhile, the remainder of the response is streamed in parallel. This streaming mechanism enables real-time interaction with the user. The module supports two operational modes, enabling seamless, natural dialog generation with Greta 2.0 :

- **Online Mode:** Online LLM inference relies on an external LLM server. Currently, Greta 2.0 supports the Mistral API, with an architecture designed to accommodate other APIs.
- **Local Mode:** Compatible with LM Studio and other locally hosted LLMs, it provides a secure, offline alternative that preserves dialog privacy.

4.2.2 Language Understanding: Dialog Act Classifier for MI. The Language Understanding module integrates a Dialog Act Classifier specifically designed for MI. This classifier assigns each user utterance to one of the given dialog acts tailored for the MI context.

The classifier is implemented using a Mistral LLM with few-shot prompting. The LLM is prompted to classify the user's utterance into one of the dialog acts from a given list (e.g., the Motivational Interview annotation schema, MISC dialog acts [27]). A definition and examples are provided in the prompt for each dialog act. The prompt includes formal definitions of each dialog act, along with examples to support accurate and consistent classification. This classifier has been evaluated in [10]. The taxonomy of dialog acts used in this module, introduced in [10], is specifically adapted for MI conversations. As defined by MISC [27], the user's possible dialog acts are *Changing Unhealthy Behavior*, *Sustaining Unhealthy Behavior*, *Sharing Negative/Positive Feelings or Emotions*, *Sharing Personal Information*, *Realization or Understanding*, *Greeting or Closing*, *Backchanneling*, and *Asking for Medical Information*.

The module can be easily adapted to other domains by providing new definitions and representative examples of dialog acts relevant to a given use case (for example, changing eating behavior, having a healthier lifestyle). The classifier can be reconfigured without any changes to the underlying architecture. The associated code and prompts are available on our project page.

4.2.3 Dialog Manager. The LLM-Dialog module also includes the Dialog Manager DREAM[12], tailored explicitly for MI. Based on the dialog act taxonomy introduced in [10], the Dialog Manager predicts

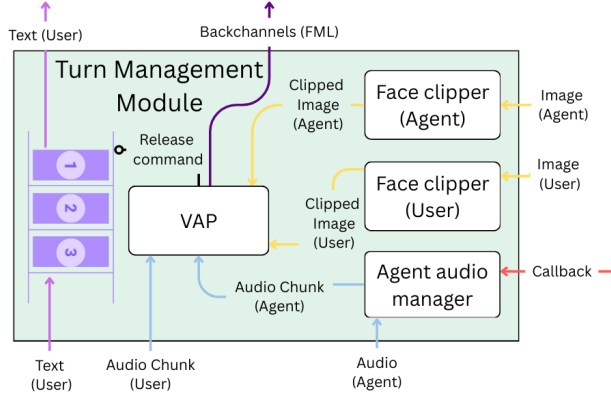


Figure 5: Dataflow of VAP-based turn-management: The module continuously receives the user’s transcript (light purple line). In parallel, the Voice Activity Projection (VAP) model predicts whether to emit agent speech responses or backchannels based on audio and clipped face image sequences from the user and the agent (blue and yellow lines). Once the backchannel timing is predicted, the predefined corresponding to the backchannel type is sent to the Behavior Planner in Functional Markup Language (FML) format (dark purple line). When the VAP model predicts agent speech response timing, the queued user’s transcripts are sent to the LLM-Dialg Module to generate the agent’s response (black line).

the next appropriate dialog act for the SIA. As defined in MISC, the agent’s possible dialog acts include *Asking for Consent or Validation*, *Providing Medical Education and Guidance*, *Planning with the Patient*, *Giving a Solution*, *Asking about Current Emotions*, *Inviting a Shift in Outlook*, *Asking for Information*, *Reflection*, *Empathic reactions*, *Acknowledging Progress and Encouraging*, *Backchanneling*, *Greeting or Closing*, and *Normalizing Experiences while Providing Reassurance*. This predicted dialog act is included in the LLM prompt, along with its definition and relevant examples. This hybrid generation approach has been validated in [11].

The Dialog Manager is implemented using a reinforcement learning (RL) framework, as described and evaluated in [9]. It follows a hierarchical RL structure comprising two levels: a master policy and several sub-policies. The master policy governs progression through the four phases of MI: Engaging, Focusing, Evoking, and Planning [27]. The sub-policies determine specific dialog strategies within each phase.

Importantly, the master policy can adapt to the three user types identified through a thorough corpus analysis, enabling dynamic and personalized conversational strategies [10].

This full ASR/LLM integration empowers Greta 2.0 to engage in fluid, speech-based interactions by leveraging LLMs’ contextual understanding and generative capabilities. The architecture also supports the integration of additional modules, such as the generation of facial expressions, which is described below [48].

While some components of this integration, such as the Natural Language Understanding module and the Dialog Manager, are designed explicitly for MI, the modular architecture allows easy

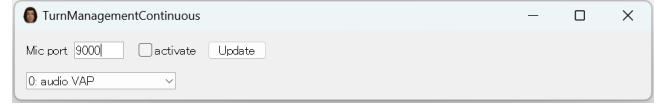


Figure 6: Interface of the turn-taking module: Once the "activate" checkbox is selected, the selected VAP model (e.g., "0: audio VAP" in the Figure) variant is loaded and starts controlling turn-taking behaviors.

adaptation to other domains by updating the prompt instructions and dialog act definitions. The prompts designed for MI are available on the project page. The dialog module was tested through a human agent studied presented in [9]

5 Turn management

Turn-taking is a fundamental phenomenon in social interactions that manages the coordinated exchange of speaking turns between participants in a conversation, which is essential for successful social communication [30, 37]. As turn-taking behaviors comprise multiple aspects (e.g., turn shift, backchannel, overlap), their modeling is a complex process.

Our turn-management module controls when the agent takes the turn by encoding both the user’s and the agent’s signals. It can utilize the different ASR module functionality modes (Section 4.1) and can manage to override the agent’s speech when the user interrupts it. Since the model predicts the start and end of the speech of the agent and the user by predicting their voice activities in parallel with the VAP window states during the model training phase, it attempts to consider the entire interaction dynamics through these predictions in inference phase, contributing to a more accurate timing prediction of the agent turn taking.

5.1 VAP-based turn-management

One recent successful example of turn-taking behavior prediction is the Voice Activity Projection model (VAP) [7]. The VAP model successfully unified multiple turn-taking behaviors (e.g., turn-shift, backchannel) with a single representation, known as the VAP window. Unlike existing turn-management models, such as those based on logistic regression of a speaker’s prosodic features [23], the VAP model thoroughly considers both speakers’ interaction dynamics from raw-level signal representations.

Although the original VAP model is trained only on audio signals of interlocutors in dyadic interactions, visual signals are also crucial for turn-taking prediction [18]. To extend the VAP capabilities, later extended VAP models incorporating other modalities were proposed, whose performance analyses demonstrated the contribution of the facial expression modality to improving the turn prediction [31, 38]. Furthermore, real-time processing speed and simplicity of requirements are also essential factors to be integrated with interaction systems. Therefore, we implemented a simplified version of the audio VAP model and a multimodal VAP model with audio and facial image signals [38].

Figure 5 shows the dataflow of the VAP-based turn-management module. The VAP models take signals from both the user and the agent as input to predict upcoming turn-taking behaviors. We used

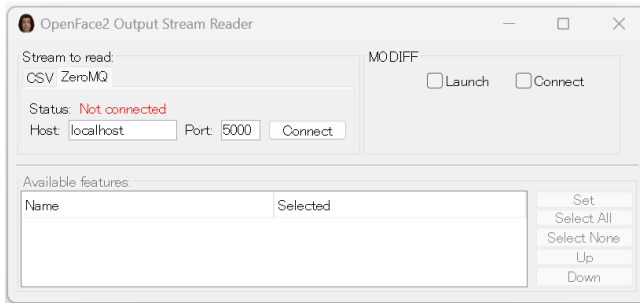


Figure 7: Interface of the OpenFace / MODIFF-8 module: "Stream to read" portion controls ZeroMQ communication with OpenFace. Once the MODIFF-8 model is activated with buttons in the "MODIFF-8" portion, AU features depicted in the "Available features" portion are loaded and start the generation system.

the VAP prediction result for SIA's turn-shift and backchannel timing. Since the Greta platform lacks an interface module (agent audio manager in Figure 5) for distributing the agent's speech audio, we control the audio transfer timing by using the callback module's signal, which indicates whether the agent is speaking. Once the agent's speech callback has been sent to this module, it begins loading the agent's audio WAV file, which was generated by the text-to-speech module. Then, it repeatedly sends small chunks to the VAP model to predict upcoming turn-taking behaviors until the speech end callback has been received. For the VAP model with audio-visual encoders, although we could directly capture the user's image from the camera, capturing and transferring the agent's image is cumbersome. Therefore, we used the agent's rendered images from the screen recorder module. These captured images are sent to the corresponding face clippers to generate face images, and then sent to the VAP model.

Every time the module receives the user's transcript text from the ASR, it puts it into a queue. Once the VAP model predicts an upcoming turn shift (from the human interlocutor to the SIA), the module releases the queued text to the LLM module, which generates the next agent utterance. In contrast, this module sends backchannels to the behavior realizer by randomly selecting pre-defined behaviors when the VAP model predicts the timing to trigger backchannels. As the VAP model cannot predict types of backchannel (e.g., supportive, competitive), we chose to use only "neutral" behaviors as the backchannel FML files corresponding to contact and perception functions [1]. The model can be switched and activated through an interface (see Figure 6).

5.2 VAD-based backchannel generation

Backchannels are short visual or vocal signals from the listener that do not interrupt the speech and are not intended to take the turn [1, 33]. Listeners can deliver their intent through the backchannels, indicating how they perceive their interlocutor's speech.

In the Greta 2.0 system, we first attempted to implement the backchannel using the aforementioned VAP model [7]. The VAP model in the original paper was trained on the audio-only Switchboard dataset. We trained it on the multimodal dataset NoXi [4],

but the result was that it generated fewer backchannels than the original version did [38]. This could be due to the limited amount of multimodal data available for NoXi compared to the audio data in the Switchboard. To overcome this issue, we implemented backchannel generation based on voice activity detection (VAD), which can run in parallel with the VAP-based turn-management module.

This module generates responses and reactive backchannels according to a few rules [3]. The response backchannel is emitted when the end of speech is detected. Specifically, the system determines the end of speech when VAD's unvoiced output lasts more than 200 ms, corresponding to an inter-pausal unit (IPU) [40]. In contrast, the system generates a reactive backchannel when VAD's voiced output lasts more than 200 ms, considering it the start of the speaker's speech. It can be interpreted by the interlocutor that the agent is paying attention to what they are saying.

6 Multimodal behavior generation (MODIFF-8)

To enable adaptive, real-time facial expression generation aligned with user behavior and dialog context, we developed the MODIFF-8 module and integrated it into Greta 2.0. This module enables the generation of socially relevant facial expressions in real-time, synchronized with the users' facial expressions and ongoing dialog acts. To develop this module, we first analyzed human-human interactions in MI settings (AnnoMI corpus [47]) and identified co-occurrences of nonverbal behaviors and MI dialog acts. We then designed an observation-to-action framework based on a conditional diffusion model, trained using the annotated sequences from the AnnoMI corpus [48]. The model conditions the generation process on a combination of multimodal inputs: the current MI dialog act, the user's facial expressions represented as action unit (AU) activations extracted using OpenFace, and turn-level temporal information (including previously expressed expressions and global interaction time). A cross-attention transformer architecture processes these inputs within the diffusion model, which learns to attend selectively to the most relevant features in the context. This allows the model to generate facial expressions for the agent that are temporally aligned with the ongoing social and dialogic dynamics. To support real-time use, MODIFF-8 employs an optimized sampling strategy based on fast ODE solvers, reducing generation latency to under 600 ms per facial expression sequence. The module is integrated in Greta 2.0 through dedicated input/output signal interfaces and GUI windows (Figure 7), allowing for the dynamic control of the OpenFace and MODIFF-8 modules.

6.1 OpenFace Integration

The module relies on real-time facial expression tracking using OpenFace 2.0, an external tool that can extract facial Action Units (AUs) from video input¹. These AUs are structured in the OpenFace CSV format, which includes frame-by-frame numeric values for a set of standardized AUs, such as AU06 (cheek raiser) and AU12 (lip corner puller), among others. The system connects to the OpenFace stream via ZeroMQ broadcasting. Incoming AU data are received by the "AU receiver" module, routed to the MODIFF-8 module (see Figure 8). The incoming signal operates at a frame rate of 25Hz, ensuring responsiveness and synchrony with the user's expressions.

¹<https://github.com/TadasBaltrusaitis/OpenFace>

6.2 MODIFF-8 Module

When the Launch checkbox from the MODIFF-8 panel of the interface is activated (see Figure 7), the system connects to the Python backend utilizing a diffusion model and initiates the real-time generation loop [48]. The MODIFF-8 module continuously receives AU sequences from the client via the OpenFace module. The images are stacked in a buffer whose size corresponds to the processing window (corresponding to the model's inference time, 600 milliseconds (15 images buffer)). The AU sequences are then processed in the MODIFF-8 input format (facial expression category sequences) using the same pipeline as that used during training. Two other inputs are sent to MODIFF-8: the extracted dialog act and the previously stored facial expression sequences (from both the human user and the agent). Then, the MODIFF-8 model produces a sequence of facial expressions for the agent for the next 600 milliseconds (15 frames). More precisely, the MODIFF-8 model is conditioned on three main input streams (see Figure 8a):

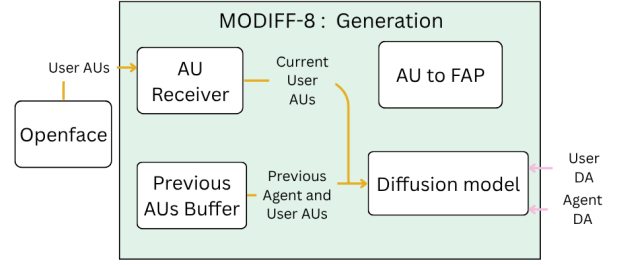
- The facial expressions of the user, captured and streamed through OpenFace ("Current User AUs" in Figure 8a).
- The dialog acts, extracted from transcripts produced by the LLM module ("User DA" and "Agent DA" in Figure 8a).
- Previously processed facial expressions of the user (respectively, previously generated facial expressions of the agent) for temporal consistency ("Previous AUs Buffer" in Figure 8b).

Figure 8 presents the flow of the MODIFF-8 Python module. It continuously receives AUs from OpenFace and dialog acts from the NLU client module (Figure 4), representing the user's current intent or the agent's ongoing response. These are aligned temporally and fed into the diffusion model, which generates adaptive AU sequences synchronized with the user's current facial expression, the dialog context, and the previously generated AUs of the agent and the processed AUs of the user stored in the "Previous AUs Buffer" (see Figure 8a). Then, the newest user AUs at time t (i.e., the "Current User AUs" in Figure 8a) and the newly generated agent AUs at time $t + 1$ are sent to the "Previous AUs Buffer" (Figure 8b). The generated AU sequences are received by the "AU to FAP" module, applying a Gaussian filter to ensure smoothness of the agent animation (see Figure 8). They are then converted into FAP frames and forwarded to the "Lip and Face Blender" module (see Figure 2), blending with other animation parameters (FAPs and BAPs).

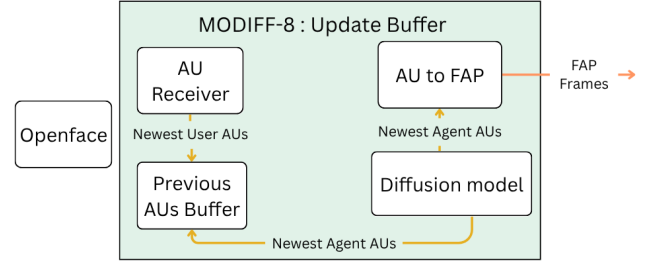
The system operates at real-time speed, generating and executing facial expressions at 25Hz (frame level). The end-to-end pipeline, from user expression extraction via AU generation to agent expression rendering, is optimized for low-latency interaction, enabling dynamic, adaptive, and socially aligned virtual agent behavior. It can generate facial expressions for the agent as a speaker and a listener. This is visible in the videos presented on the website <https://youtu.be/uzi6gprzN3c>. This MODIFF-8 module was validated in an experimental study where human participants interacted with the virtual agent in three conditions: only lip shapes, mismatch condition where the agent displayed behaviors asynchronously as those of its human interlocutor, and our model [49].

7 Conclusion

This paper introduced the Greta 2.0 system, which extends the SAIBA-based Greta platform to include neural network models. By



(a) **Generation step:** The user's AUs are concatenated with the previous AU buffer and sent to the Diffusion model. It also receives the user's and the agent's Dialog Acts (DAs) to generate the next agent's AUs.



(b) **Buffer update step:** While the generated agent AUs are converted into the Facial Animation Parameter (FAP) frames, the Previous AUs Buffer is updated with the user's AUs and the generated agent's AUs.

Figure 8: MODIFF-8 module dataflow

bypassing the SAIBA process with direct frame-level modification while synchronizing them, the system integrates frame-by-frame generation models such as text-based LLMs, end-to-end ASR models, and turn-taking prediction models. While the integration of neural networks significantly enhances the system's adaptive interactions, it also introduces higher computational demands compared to the original SAIBA-based Greta platform, such as a 32GB GPU. This extended version of the Greta system still employs a modular-based design; thus, individual components can be replaced within the interaction pipeline, enabling new models to be tested. In addition to this flexible design, the system is open-source under the GNU GPL v3 license and can be deployed on any compatible machine. To date, the MODIFF-8 and dialog manager modules have been evaluated only within the framework of MI. They should be applied to other scenarios to evaluate their generalizability. While the platform can stop the agent's speech when the user interrupts it, it has not yet incorporated the replanning of the agent's verbal and nonverbal behavior. Nevertheless, the current system provides a strong foundation for future improvements, making it straightforward to extend with additional functionalities, gesture generation, or support for group dialogues.

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